

Tyler Chandross-Cohen

Curriculum Vitae

Ph.D. Candidate, The Pennsylvania State University
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113 Alma Mater Dr, Apt 303, State College PA, 16803

SUMMARY

Food microbiology researcher with a strong record of conducting interdisciplinary research at the interface of microbial genomics, molecular biology, and food safety. Research centers on the foodborne pathogen *Bacillus cereus*, with emphasis on how genetic diversity and environmental conditions govern virulence expression and cytotoxic potential across diverse isolates, including psychrotolerant strains relevant to refrigerated foods. Integrates whole-genome sequencing with in vitro human cell-based cytotoxicity models to assess the virulence potential of *B. cereus* isolates originating from diverse environmental and food-associated sources. Employs integrated genomic, transcriptional, and phenotypic approaches to characterize *B. cereus* under food-relevant conditions, linking molecular signatures with functional outcomes in human intestinal epithelial cell models. In parallel with research, actively engaged in mentoring undergraduate researchers and teaching.

EDUCATION AND TRAINING

The Pennsylvania State University

Ph.D. Candidate, Food Science

Aug 2022 – Present

GPA: 3.98

Relevant coursework: Applied Bioinformatics (852), Applied Statistics (500), Professional Skills for Graduate Students (503), Research Methods (501).

The Pennsylvania State University

B.S., Food Science (Minor: Microbiology)

2022

GPA: 3.68

Relevant coursework: Food Microbiology (408), Food Microbiology Lab (409), Communicating Research in Agricultural Sciences (422), Food Engineering (405), Food Law (497), Medical Microbiology (412), Microbial Genomic Epidemiology (517).

Dean's List: Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, Spring 2022

RESEARCH EXPERIENCE

The Pennsylvania State University, Food Science Department

Graduate Research Assistant (Advisor: Dr. Jasna Kovac)

Aug 2022 – Present

- Conduct doctoral research on the foodborne pathogen *Bacillus cereus*, with emphasis on psychrotolerant isolates and targeted gene knockout strategies to better understand virulence gene-mediated cytotoxicity.
- Design and perform growth experiments in skim milk broth across multiple temperature schemes to inform quantitative exposure assessment models.
- Quantify cytotoxicity of *B. cereus* isolates using human intestinal epithelial cell culture models to evaluate virulence potential under food-relevant conditions.
- Train and mentor undergraduate researchers; oversee laboratory operations including media preparation; and develop detailed laboratory protocols, standard operating procedures (SOPs), and experimental proposals.

The Pennsylvania State University, Food Science Department

Undergraduate Research Associate (Advisor: Dr. Jasna Kovac)

Jan 2021 – May 2022

- Collaborated with a graduate student researcher to detect and quantify *Listeria monocytogenes* from environmental samples collected in ice cream manufacturing facilities.
- Isolated and characterized protective cultures and evaluated their inhibitory activity against *L. monocytogenes* using standard microbiological and analytical assays.

The Pennsylvania State University, Food Science Department

Undergraduate Research Assistant (Advisor: Dr. Molly Kelly)

Oct 2018 – May 2022

- Conducted undergraduate research focusing on experimental evaluation of the impact of the spotted lanternfly on wine quality and safety.
- Designed and performed experiments to assess effects of spotted lanternfly infestation on wine fermentation, chemical composition, and overall product quality.

- Analyzed yeast strain performance with respect to aroma and flavor development through controlled fermentation experiments and sensory-related analytical assays.
- Monitored daily wine production activities, including systematic sampling, chemical testing, and quantitative data collection and analysis.

The Pennsylvania State University, Plant Science Department
Undergraduate Research Assistant (Advisor: Dr. Michela Centinari)

Sept 2019 – Mar 2020

- Assisted with research projects supporting viticulture and enology-focused experimental studies.
- Conducted experiments evaluating the effects of budburst-delay strategies in grapevines on grape composition and downstream wine chemistry.
- Performed berry sampling, grape juice chemical analyses, and carbohydrate profiling; managed data entry; assisted with grape processing; and monitored wine fermentations.

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Michel, P., Maxson, T., Chivukula, V., Overholt, W., Medina Cordoba, L. K., Ayodele-Abiola, S., McQuiston, J., Beesley, C. A., Bell, M., Caban Figueroa, V., Bugrysheva, J., Chandross-Cohen, T., Weiner, Z., Carroll, L., Kovac, J., & Sue, D. (2026). Antimicrobial resistance profiling and phenotypic characterization of archived clinical *Bacillus paranthracis* strains. *Microbiology Spectrum*, 8(8), e00875-26. DOI: 10.1128/spectrum.00875-26.
Contribution: Conducted experiments, writing review & editing.
2. Chandross-Cohen, T., Yount, M., Readinger, E., Prince, C., Kimble, K., Praul, B., Weidner, L., & Kovac, J. (2026). Environmental Modulation of Cytotoxicity and Virulence Factor Stability in Psychrotolerant *Bacillus cereus* Group Isolates. *Food Microbiology*. DOI: 10.1016/j.fm.2026.105166.
Contribution: Writing original draft, writing review & editing, investigation, formal analysis, visualization, data curation, conceptualization.
3. Chandross-Cohen, T., Ebersole, M., Rahman, N., Wagner, A. S., Kovac, J., & Wildschutte, H. (2026). Whole-genome Sequence and Phenotypic Characteristics of *Bacillus cereus* Strain IM8 Isolated from the Upper Fremont Glacier in Wyoming. *Microbiology Resource Announcements*. DOI: 10.1128/mra.01260-25.
Contribution: Conducted experiments, writing original draft, writing review & editing, visualization.
4. Nguyen, P., Chandross-Cohen, T., Kovac, J., & Restaino, L. (2025). Isolation and Characterization of *Bacillus cereus* Group Species in Powdered Infant Formula and Infant Cereal Using a Newly Developed Detection System. *Journal of Food Protection*. DOI: 10.1016/j.jfp.2025.100677.
Contribution: Conducted experiments, writing original draft, writing review & editing, visualization.
5. Navarre, A., Rupert, K., Chandross-Cohen, T., & Kovac, J. (2025). Low Prevalence and Concentrations of *Campylobacter* Detected on Retail Chicken Breasts. *Journal of Food Protection*. DOI: 10.1016/j.jfp.2025.100635.
Contribution: Whole genome sequencing of isolates, writing review & editing.
6. Chandross-Cohen, T., Chung, T., Watson, S., Rolon, M. L., & Kovac, J. (2025). Precision Food Safety: Advances in Omics-based Surveillance for Proactive Detection and Management of Foodborne Pathogens. *Trends in Food Science & Technology*, 105186. DOI: 10.1016/j.tifs.2025.105186.
Contribution: Writing original draft, writing review & editing, visualization.
7. Su, J.*, Chandross-Cohen, T.*, Qian, C., Carroll, L., Kimble, K., Yount, M., Wiedmann, M., & Kovac, J. (2024). Assessment of the Exposure to Cytotoxic *Bacillus cereus* Group Genotypes Through HTST Milk Consumption. *Journal of Dairy Science*. DOI: 10.3168/jds.2024-24703. (*Co-first authors)
Contribution: Development of methods, conducted experiments, data analysis, and co-wrote and revised the manuscript.
8. Wei, X., Hassen, A., McWilliams, K., Peitrzen, K., Chung, T., Mendez Acevedo, M., Chandross-Cohen, T., Dudley, E. G., Vipham, J., Mamo, H., Sisay Tessema, T., Zewdu, A., & Kovac, J. (2024). Genomic Characterization of *Listeria monocytogenes* and *Listeria innocua* Isolated from Milk and Dairy Samples in Ethiopia. *BMC Genomic Data*, 25, 12. DOI: 10.1186/s12863-024-01195-0.
Contribution: Re-confirmed the Listeria spp. at The Pennsylvania State University.
9. Rolon, M. L., Chandross-Cohen, T., Kaylegian, K. E., Roberts, R. F., & Kovac, J. (2023). Context Matters: Environmental Microbiota of Ice Cream Processing Facilities Affects the Inhibitory Performance of Two Lactic Acid Bacteria Against *Listeria monocytogenes*. *ASM Microbiology Spectrum*, 12(1). DOI: 10.1128/spectrum.01167-23.
Contribution: Conducted experiments, conducted data analysis, and co-wrote and revised the manuscript.

Publications Under Review or In Preparation

1. Prince, C., Chandross-Cohen, T., Chung, T., Kimble, K., Wiedmann, M., Johler, S., & Kovac, J. (2026). Single nucleotide polymorphisms associated with cytotoxicity of *Bacillus cereus* group strains in Caco-2

cells. *mSphere*, in revision.

Contribution: Conducted experiments, writing review & editing.

2. Su, J., Chandross-Cohen, T., Qian, C., Carroll, L., Kovac, J., Ivanek, R., & Wiedmann, M. (2026). Development of BRisk: a digital tool for food safety risk assessment of *Bacillus cereus* group strains in HTST pasteurized fluid milk. *Journal of Dairy Science*, in review.

Contribution: Conducted experiments, writing review & editing.

3. Chandross-Cohen, T., Weidner, L., Fratnik Steyer, A., Petrovic, Z., & Kovac, J. (2026). Validation of SNPs associated with cytotoxicity in and around enterotoxin genes of *Bacillus cereus* using food, environmental, and human clinical isolates. In preparation.

Contribution: Writing original draft, writing review & editing, investigation, formal analysis, visualization, data curation, conceptualization.

4. Vieira, Y. C., Chandross-Cohen, T., & Kovac, J. (2026). Whole-genome amplification and targeted CRISPR-Cas9 enrichment enables environmental *Staphylococcus aureus* detection in low biomass samples. In preparation.

Contribution: Writing review & editing, investigation, data curation, conceptualization.

5. Wei, X., Chung, T., Chandross-Cohen, T., Miller, G. H., & Kovac, J. (2026). Detection and characterization of Bacilli plasmidome. In preparation.

Contribution: Conducted experiments, writing review & editing.

PRESENTATIONS

Oral Presentations

1. Chandross-Cohen, T., Nguyen, P. T., Restaino, L., & Kovac, J. (2026). Cereus-ly a Concern: Genomic and Cytotoxic Characterization of *Bacillus cereus* Group Isolates from Infant Foods. Pennsylvania Association for Food Protection Annual Meeting, May 14–15, 2026, University Park, PA. *Invited*.

*Contribution: Development of methods, conducted experiments, conducted data analysis, and **presented**.*

2. Chandross-Cohen, T., Yount, M., Readinger, E., Prince, C., Kimble, K., Praul, B., Centeno, C., & Kovac, J. (2025). Cytotoxicity, Transcription, and Stability of Virulence Factors Produced by Psychrotolerant *Bacillus cereus* Group Isolates. International Association for Food Protection Annual Meeting, July 28–30, 2025, Cleveland, OH.

*Contribution: Development of methods, conducted experiments, conducted data analysis, and **presented**.*

Poster Presentations

1. Chandross-Cohen, T., Weidner, L., Fratnik Steyer, A., Petrovic, Z., & Kovac, J. (2026). Genotypic and Phenotypic Characterization of *Bacillus cereus* Isolates from Human Clinical Cases in Slovenia. International ICFMH Conference FOODMicro2026, Sept 7–10, 2026, Ljubljana, Slovenia.

*Contribution: Development of methods, conducted experiments, conducted data analysis, and **presented**.*

2. Chandross-Cohen, T., Weidner, L., Fratnik Steyer, A., Petrovic, Z., & Kovac, J. (2026). Genotypic and Phenotypic Characterization of *Bacillus cereus* Isolates from Human Clinical Cases in Slovenia. International Association for Food Protection Annual Meeting, July 27–29, 2026, New Orleans, LA.

*Contribution: Development of methods, conducted experiments, conducted data analysis, and **presented**.*

3. Chandross-Cohen, T., Nguyen, P. T., Restaino, L., & Kovac, J. (2026). Characterization of *Bacillus cereus* Group Isolates from Infant Formula and Cereal by Whole-Genome Sequencing and Cytotoxicity. ASM Microbe, June 4–8, 2026, Washington, DC.

*Contribution: Development of methods, conducted experiments, conducted data analysis, and **presented**.*

4. Cavalcante, Y., Chandross-Cohen, T., Mouratidis, I., Patsakis, M., Georgakopoulos-Soares, I., & Kovac, J. (2026). Culture-independent Detection of *Staphylococcus aureus* Using Multiple Displacement Amplification and CRISPR-Cas9 Coupled with Nanopore Sequencing. One Health Microbiome Symposium, Penn State One Health Microbiome Center, May 14, 2026, University Park, PA.

Contribution: Development of methods, conducted experiments.

5. Chandross-Cohen, T., Nguyen, P. T., Restaino, L., & Kovac, J. (2026). Characterization of *Bacillus cereus* Group Isolates from Infant Formula and Cereal by Whole-Genome Sequencing and Cytotoxicity. Graduate Exhibition, Mar 27, 2026, University Park, PA.

*Contribution: Development of methods, conducted experiments, conducted data analysis, and **presented**.*

6. Cavalcante, Y., Chandross-Cohen, T., & Kovac, J. (2026). Culture-independent Detection of *Staphylococcus aureus* Using Multiple Displacement Amplification and CRISPR-Cas9 Coupled with Nanopore Sequencing. Graduate Exhibition, Mar 27, 2026, University Park, PA.

Contribution: Development of methods, conducted experiments, conducted data analysis.

7. Chandross-Cohen, T., Nguyen, P. T., Restaino, L., & Kovac, J. (2026). Characterization of *Bacillus cereus*

- Group Isolates from Infant Formula and Cereal by Whole-Genome Sequencing and Cytotoxicity. Gamma Sigma Delta Symposium, Mar 26, 2026, University Park, PA. *Poster Award.*
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
8. Cavalcante, Y., Chandross-Cohen, T., & Kovac, J. (2026). Culture-independent Detection of *Staphylococcus aureus* Using Multiple Displacement Amplification and CRISPR-Cas9 Coupled with Nanopore Sequencing. Gamma Sigma Delta Symposium, Mar 26, 2026, University Park, PA.
Contribution: Development of methods, conducted experiments, conducted data analysis.
 9. Weidner, L., Chandross-Cohen, T., Fratik Steyer, A., Petrovic, Z., & Kovac, J. (2026). Genotypic and Phenotypic Characterization of *Bacillus cereus* Isolates from Human Clinical Cases in Slovenia. Gamma Sigma Delta Symposium, Mar 26, 2026, University Park, PA. *Poster Award.*
Contribution: Development of methods, conducted experiments, conducted data analysis.
 10. Lyons, M., Biernbaum, E., Chandross-Cohen, T., Cavalcante, Y., Kovac, J., & Dudley, E. (2025). Targeted Nanopore Sequencing of the O-Antigen Region in *E. coli* Using CRISPR Cleavage at *gndA*. Allegheny Branch of the American Society of Microbiology Annual Meeting, Nov 7–8, 2025, California, PA.
Contribution: Development of methods, conducted experiments, conducted data analysis.
 11. Chandross-Cohen, T., Praul, B., & Kovac, J. (2025). Effects of Protein Stress Factors on the Cytotoxicity of Supernatants from Psychrotolerant *Bacillus cereus* Isolates. IAFP European Symposium, May 6–8, 2025, Madrid, Spain.
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
 12. Chandross-Cohen, T., Praul, B., & Kovac, J. (2025). Effects of Protein Stress Factors on the Cytotoxicity of Supernatants from Psychrotolerant *Bacillus cereus* Isolates. Gamma Sigma Delta Symposium, Mar 27, 2025, University Park, PA. *Poster Award.*
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
 13. Chandross-Cohen, T., Yount, M., Readinger, E., Prince, C., Kimble, K., Centeno, C., & Kovac, J. (2024). Cytotoxicity Assessment of Psychrotolerant *Bacillus cereus* Isolates Across Varied Temperatures. Allegheny Branch of the American Society of Microbiology Annual Meeting, Nov 1–2, 2024, Greensburg, PA.
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
 14. Praul, B., Chandross-Cohen, T., & Kovac, J. (2024). Effects of Protein Stress Factors on the Cytotoxicity of Psychrotolerant *Bacillus cereus* Isolates. Allegheny Branch of the American Society of Microbiology Annual Meeting, Nov 1–2, 2024, Greensburg, PA.
Contribution: Development of methods, conducted experiments, conducted data analysis.
 15. Chandross-Cohen, T., Yount, M., Readinger, E., Prince, C., Kimble, K., Centeno, C., & Kovac, J. (2024). Cytotoxicity Assessment of Psychrotolerant *Bacillus cereus* Isolates Across Varied Temperatures. International Association for Food Protection Annual Meeting, July 15–17, 2024, Long Beach, CA.
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
 16. Chandross-Cohen, T., & Kovac, J. (2024). Variation in *Bacillus cereus* Hemolysin BL Transcription Associated with a SNP in the Transcription Regulatory Region. Gamma Sigma Delta Symposium, Mar 27, 2024, University Park, PA.
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
 17. Yount, M., Chandross-Cohen, T., Georgiana, G., Readinger, E., & Kovac, J. (2024). Surface Colonization Area Was Associated with Phylogenetic Group and Hemolysin BL Production in *Bacillus cereus* sensu lato. Gamma Sigma Delta Symposium, Mar 27, 2024, University Park, PA. *Poster Award.*
Contribution: Development of methods, assistance with experiments, conducted data analysis.
 18. Chandross-Cohen, T., & Kovac, J. (2024). Variation in *Bacillus cereus* Hemolysin BL Transcription Associated with a SNP in the Transcription Regulatory Region. Food Science in Action Symposium, Mar 20, 2024, University Park, PA.
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
 19. Chandross-Cohen, T., Yount, M., Su, J., Qian, C., Wiedmann, M., & Kovac, J. (2023). Growth Potential of *Bacillus cereus* Group Strains from Different Phylogenetic Groups in a Dairy Food Model. Allegheny Branch of the American Society of Microbiology Annual Meeting, Nov 3–4, 2023, Williamsport, PA.
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
 20. Chandross-Cohen, T., Yount, M., Su, J., Qian, C., Wiedmann, M., & Kovac, J. (2023). Growth Potential of *Bacillus cereus* Group Strains from Different Phylogenetic Groups in a Dairy Food Model. International Association for Food Protection Annual Meeting, July 16–19, 2023, Toronto, ON, Canada.
Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.
 21. Su, J., Qian, C., Chandross-Cohen, T., Yount, M., Wiedmann, M., & Kovac, J. (2023). An Exposure Assessment of Cytotoxic *Bacillus cereus* Strains from Various Phylogenetic Groups in HTST Milk. International Association for Food Protection Annual Meeting, July 16–19, 2023, Toronto, ON, Canada.
Contribution: Provided data used in growth modeling.
 22. Chandross-Cohen, T., Yount, M., Su, J., Qian, C., Wiedmann, M., & Kovac, J. (2023). Growth Potential

of *Bacillus cereus* Group Strains from Different Phylogenetic Groups in a Dairy Food Model. Food Science in Action Symposium, May 26, 2023, University Park, PA.

Contribution: Development of methods, conducted experiments, conducted data analysis, and co-presented.

23. Chandross-Cohen, T., Yount, M., Su, J., Qian, C., Wiedmann, M., & Kovac, J. (2023). Growth Potential of *Bacillus cereus* Group Strains from Different Phylogenetic Groups in a Dairy Food Model. Gamma Sigma Delta Symposium, Mar 30, 2023, University Park, PA. *Poster Award.*

Contribution: Development of methods, conducted experiments, conducted data analysis, and presented.

24. Rolon, M. L., Chandross-Cohen, T., Kaylegian, K. E., Roberts, R. F., & Kovac, J. (2022). Context Matters: Environmental Microbiota of Ice Cream Processing Facilities Affects the Inhibitory Performance of Two Lactic Acid Bacteria Against *Listeria monocytogenes*. Allegheny Branch of the American Society of Microbiology Annual Meeting, Nov 4–5, 2022, Morgantown, WV.

Contribution: Conducted experiments, conducted data analysis, and presented.

25. Rolon, M. L., Chandross-Cohen, T., Kaylegian, K. E., Roberts, R. F., & Kovac, J. (2022). Context Matters: Environmental Microbiota of Ice Cream Processing Facilities Affects the Inhibitory Performance of Two Lactic Acid Bacteria Against *Listeria monocytogenes*. International Association for Food Protection Annual Meeting, July 31–Aug 3, 2022, Pittsburgh, PA.

Contribution: Conducted experiments and conducted data analysis.

26. Chandross-Cohen, T., Rolon, M. L., & Kovac, J. (2022). Isolation, Characterization, and Application of Antilisterial Isolates in a Raw Milk Cheese Model to Inhibit *Listeria monocytogenes*. Gamma Sigma Delta Symposium, Mar 31, 2022, University Park, PA. *Poster Award.*

Contribution: Original ideas, conducted experiments, conducted data analysis, and presented.

TEACHING AND ADVISING

Courses / Instruction

The Pennsylvania State University, Food Science Department

Instructor on Record, Food Microbiology Laboratory (FDSC 409)

Fall 2025

- Taught two laboratory sections per day, twice weekly (Tuesdays and Thursdays), providing instruction to undergraduate students on microbiological methods in food safety and quality.
- Delivered pre-lab lectures and guided students through exercises involving isolation, enumeration, and identification of foodborne pathogens and spoilage microorganisms.
- Implemented and maintained pre-established course curriculum, protocols, and assessments.
- Supervised and mentored students in aseptic technique, serial dilutions, plate counting, enrichment and selective culturing, and biochemical characterization.
- Managed biosafety and laboratory operations, ensuring compliance with BSL-2 practices and appropriate waste handling.
- Evaluated student progress through lab reports, quizzes, and practical exams.

The Pennsylvania State University, Food Science Department

Short Course Graduate Teaching Assistant, Ice Cream Short Course

Spring 2023, 2024, 2025, 2026

- Assisted in laboratory sessions on ice cream freezing, mix calculation, and vanilla flavor.
- Assisted in homework help sessions to ensure the success of the students.

The Pennsylvania State University, Food Science Department

Graduate Teaching Assistant, Food Microbiology Laboratory (FDSC 409)

Fall 2023, Fall 2024

- Present in class to assist in class discussions and troubleshoot students' experiments.
- Hosted weekly office hours to guide students and answer their questions.
- Graded laboratory reports, results sheets, and exams.

The Pennsylvania State University, Chemistry Department

Learning Assistant and Grader

Fall 2020, Spring 2021, Fall 2021, Spring 2022

- Hosted two weekly LA sessions to review problem sets with students.
- Supported outside-of-class learning and gave extra help to students.
- Graded midterm and final exams.

The Pennsylvania State University, Food Science Department

Undergraduate Teaching Assistant, Food Microbiology (FDSC 408)

Fall 2021

- Present in class to answer questions and involved in class discussions.
- Hosted weekly office hours to answer students' questions and support their learning; reviewed and proctored exams.

Mentorship

• Lydia Weidner [†]	Nov 2024 – Current
• Andrew Posey [†]	April 2026 – May 2026
• Brian Praul [†]	July 2024 – Aug 2024
• Kailee Shotto [†]	Aug 2023 – May 2024
• Elizabeth Henry [†]	Aug 2023 – Jan 2024
• Erin Readinger [†]	Jan 2023 – May 2024
• Mackenna Yount [†]	Aug 2022 – May 2024
• Express Mentorship Group, Rutgers Preparatory School	Aug 2021 – Current

[†]Undergraduate student at Penn State

GRANTS & COMPETITIVE FUNDING

External Grants

Funded

- USDA NIFA Rapid Response (2026–2028). *Prevention of Cereulide Contamination in Infant Formula Through Enhanced Bacillus cereus Detection and Risk Assessment*. Lead PI: Jasna Kovac. Role: Assisted in grant development. Amount: \$500,000. Credit: 0%.
- USDA NIFA Predoctoral Fellowship (2024–2027). *Assessing The Role of Hemolysin BL in Cytotoxicity and Fitness of B. cereus in Food Systems*.[#] Role: PI. Amount: \$180,000. Credit: 100%.

Not Funded

- NSF Graduate Research Fellowship Program (GRFP) (2023–2026). *Characterizing the Ecological Role of B. cereus Hemolysin BL in biofilm formation and competition*. Role: PI. Amount: \$160,000. Credit: 100%.

Internal Grants

Funded

- Penn State College of Agricultural Sciences (2023–2024). *Assessing the Effect of Polymorphisms in hbl Genes in B. cereus on Transcription of hbl and Cytotoxicity*.[#] Role: PI. Amount: \$3,000. Credit: 100%.
- Penn State Department of Food Science (2021). *Isolation, Characterization, and the Use of Protective Cultures in a Raw Milk Cheese Model to Inhibit Listeria monocytogenes*.^{*} Role: PI. Amount: \$6,000. Credit: 100%.

Not Funded

- Penn State College of Agricultural Sciences (2021–2022). *Evaluating the Antilisterial Activity of Protective Cultures in Food Processing Environmental in vitro Biofilms*.^{*} Role: PI. Amount: \$2,000. Credit: 100%.

^{*}Undergraduate research funding; [#]Graduate research funding

LEADERSHIP EXPERIENCE

• Kovac Lab Safety Coordinator	Jan 2023 – Current
• Treasurer, IAFP Student Professional Development Group	May 2023 – July 2025
• President, Penn State Chapter of Phi Tau Sigma	Apr 2023 – May 2026
• Treasurer, Penn State Chapter of Phi Tau Sigma	May 2022 – Apr 2023

AWARDS & HONORS

• Penn State Graduate Student International Travel Grant [*]	2026
• Gamma Sigma Delta Symposium, First Place Winner, Food Systems Graduate Division [*]	2026
• Gamma Sigma Delta Symposium, First Place Winner, Microbiology/Microbiome Graduate Division [*]	2025
• Dr. Daryl B. & Mrs. Dawn L. Lund Student International Travel Scholarship of Phi Tau Sigma [°]	2025
• WH Pierce Prize for Global Impact in Microbiology, One Health Microbiome Center, presented by Applied Microbiology International [*]	2024
• John H Hetrick Food Science Scholarship [*]	2024
• International Association for Food Protection Student Travel Award [°]	2024

• Allegheny Branch, American Society of Microbiology Meeting, Second Place Winner, Graduate Division Poster Presentation [◊]	2023
• Dudek Graduate Scholarship in Food Science*	2023
• Internal Agricultural Sciences Scholarship*	2023
• Food Industry Group Graduate Student Leadership Award*	2023
• Gamma Sigma Delta Symposium, First Place Winner, Graduate Division*	2023
• Internal Agricultural Sciences Scholarship*	2023
• PNC Financial Corporation Graduate Fellowship*	2023
• Robert Graham Endowment Graduate Fellowship*	2022
• Keeney Food Science Department Head Fund*	2022
• University Graduate Fellowship*	2022
• E&W Rosskam Scholarship in Food Science*	2022
• Henry Pierce in Ag Science Scholarship*	2022
• Gamma Sigma Delta Symposium, First Place Winner, Undergraduate Division*	2022
• Virginia Dare Award [◊]	2021
• Zepp Family Food Safety Endowment*	2021
• Henry Pierce in Ag Science Scholarship*	2021
• American Society for Enology and Viticulture National Scholarship [◊]	2021
• Featured Author, Penn Statements Volume 39, Spring 2020 Edition*	2020
• Zepp Family Food Safety Endowment*	2020
• Keystone Institute of Food Technologists Scholarship [◊]	2020
• Henry Pierce in Ag Science Scholarship*	2020
• American Society for Enology and Viticulture National Scholarship [◊]	2020
• National Association of Flavors & Food Ingredient Systems Scholarship Award [◊]	2019
• Featured Student, Penn State College of Ag Sciences “Ag Student Journeys”*	2019

*Penn State affiliated scholarship/fellowship/award; [◊]External scholarship/fellowship/award

CERTIFICATIONS

• CITI OSHA Bloodborne Pathogens	2023 – Current
• Hazard Analysis and Critical Control Points (HACCP)	2022 – Current
• CITI Biosafety and Biosecurity (BSS)	2019 – Current
• Laboratory and Research Training	2018 – Current

PROFESSIONAL MEMBERSHIPS

• International Association for Food Protection (IAFP)	2022 – Current
• Penn State One Health Microbiome Center	2022 – Current
• American Society for Microbiology (ASM)	2022 – Current
• Allegheny Branch of the American Society for Microbiology (ABASM)	2022 – Current
• Penn State Chapter of Phi Tau Sigma	2021 – Current
• Phi Tau Sigma National Honors Society	2021 – Current
• Institute of Food Technologists (IFT)	2018 – Current
• Penn State Food Science Club	2018 – 2022